Code No: 113BS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, August/September - 2022
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain BCD codes, Excess-3 code, Gray code, and ASCII codes used in digital systems.
b) Write about error detecting codes and error correcting codes used in digital systems.
c) Convert (i) $68BE_{16}$ to Binary and then Octal (ii) Convert 345_{10} to binary and hexadecimal. [5+5+5]

2.a) Explain the axiomatic definition of Boolean algebra by taking two and three variables.
b) Explain the characteristics of the logic families (i) TTL (ii) ECL (iii) MOS (iv) CMOS. [7+8]

3.a) Explain the methods of simplification of Boolean expression using Karnaugh methods by taking 4 variable function that includes Don't care conditions.
b) Simplify the following four variable Boolean function and implement the same using NAND logic. $F(A, B, C, D) = \sum m(0, 2, 4, 5, 7, 8, 10, 13, 15), \sum d(3,6,11,12)$. [8+7]

4.a) Minimize the following multiple output functions using SOP:
$f_1 = \sum m(0,1,2,4,6,7,10)$
$f_2 = \sum m(2,4,10,12,14)$
b) Explain the construction of Karnaugh's maps with five variables and simplify the Boolean function $F(A,B,C,D, E) = \sum m(0,1,2,4,5,7,11,15, 18, 20, 23, 25,26,27, 30)$. [7+8]

5.a) Discuss the analysis and design procedure and draw the logic diagram of a 4 bit BCD adder/subtractor.
b) Explain functions of Multiplexers and Decoders. Design combinational circuits for 4/1 line multiplexers using logic gates. [7+8]

6.a) Construct a 4 bit 2's complement adder and a BCD adder using full adders and perform addition and subtraction by taking 4-bit numbers with examples.
b) Draw the logic diagrams of 4 bit magnitude comparator and explain its working with typical examples. [7+8]

7.a) Draw the logic diagram of a 4-bit binary ripple counter with parallel load and explain its working with its function table.
b) Design a 4 bit synchronous counter with JK flip - flops and explain its working with the help of state diagrams and state tables. [7+8]

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8.a) Given 32 × 8 ROM with enable input, Show the external connections necessary to construct a 128 × 8 ROM with 4 chips and a decoder.

b) What are PLAs? Explain the working of a PLA with a schematic and implement the following two Boolean functions with a PLA:
$F_1 (A, B, C) = \sum (0, 1, 2, 4)$ and $F_2 (A, B, C) = \sum (0, 5, 6, 7)$. [7+8]

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